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Dynamic Multiscale DLinear for Long-Horizon Time-Series Forecasting

A Time Series Analysis Final Project

Course: Time Series Analysis (361.2.2480)

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Selected paper: "Are Transformers Effective for Time Series Forecasting?"

https://github.com/Etaiwi/long-horizon-time-series-forecasting

1. Introduction and Project Goals

In this project, we studied long-horizon time-series forecasting through DLinear, a simple model introduced by Zeng et al. [1] as an alternative to much larger Transformer-based forecasting models. DLinear is especially useful for this project because its architecture is easy to interpret. It separates each input series into a smooth trend and a remainder, then forecasts both components with direct linear projections.

Our first goal was to reconstruct the published DLinear experiment on the public Weather dataset and verify that our implementation reproduces the paper's reported errors. Our second goal was to introduce one meaningful improvement while keeping the original model recognizable and computationally light. The question that guided the improvement was: can DLinear benefit from choosing its trend scale according to both the measured variable and the current input window?

1.1 Selected Paper and Reconstruction Target

The selected paper, *Are Transformers Effective for Time Series Forecasting?*, introduces the LTSF-Linear family and evaluates it on nine public time-series benchmarks [1]. We focus on DLinear and reconstruct its multivariate Weather experiment with input length 336 and forecast horizons 96, 192, 336, and 720. The official implementation and experiment scripts are publicly available [2], and the benchmark data are linked by the authors [3].

1.2 Project Scope and Contributions

The submitted project contains four executed notebooks. Together, they:

1. Build a reproducible, leakage-safe Weather data pipeline and seasonal-naive baseline.
2. Reconstruct the original DLinear model and compare it directly with the paper.
3. Introduce static and dynamic multiscale versions of the decomposition.
4. Evaluate the final improvement with paired random seeds and all four paper horizons.

The main reconstruction uses the paper's seed 2021. For the improvement, we additionally use seeds 2022 and 2023 so that the conclusion does not depend on one favorable initialization.

2. Dataset and Experimental Setup

2.1 Forecasting Task

The task is supervised because every training input window is paired with its known future target window and the model parameters are optimized to reduce forecast error. It is multivariate-to-multivariate because all 21 historical Weather variables are provided, and all 21 future variables are predicted.

The standardized input X has shape $[B, L, C]$, where B is the batch size, $L = 336$ is the input length, and $C = 21$ is the number of variables. The target Y has shape $[B, H, C]$, where the main forecast horizon is $H = 96$. Weather is sampled every ten minutes, so the input covers 56 hours, and the main output covers 16 hours.

DLinear performs direct multi-step forecasting: it predicts the complete output window in one forward pass. It does not forecast one point and feed that prediction recursively into the next step.

2.2 Weather Dataset and Data-Quality Decisions

The public Weather file contains 52,696 chronologically ordered rows, one datetime column, and 21 numeric variables. During loading, we parse the date column as datetimes and verify that all measurement values are numeric, finite, and non-missing. No resampling or value imputation is applied, preserving the official benchmark row sequence.

The file contains one exact duplicated row at 2020-05-12 06:00 and one 100-minute gap at 2020-05-29, corresponding to nine missing ten-minute timestamps. Both irregularities are inside the training partition. We retain them to match the official row-based benchmark and do not interpolate the gap.

This creates a small irregular-spacing limitation, but it does not create leakage from validation or test into training.

2.3 Temporal Split and Leakage Prevention

We preserve the official row order and use the paper's chronological 70%/10%/20% train/validation/test split. A standard scaler is fitted only on the training partition and then applied unchanged to validation and test.

Input-target windows are constructed so that each target remains completely inside its assigned partition. Validation and test inputs may include the observations immediately before the target partition because those historical observations would be known at forecast time. Hyperparameters and model selection use validation MSE; test targets are evaluated only after the configuration is fixed.

2.4 Metrics and Seasonal-Naive Baseline

The paper reports mean squared error (MSE) and mean absolute error (MAE), where lower values are better. MSE gives more weight to large errors, while MAE averages absolute deviations and is less sensitive to unusually large mistakes. All metrics are aggregated across every test window, forecast step, and variable in standardized space.

Our simple baseline is daily seasonal naive. One day contains 144 ten-minute samples, so the baseline repeats the most recent daily pattern into the forecast horizon. At horizon 96 it obtains:

$$\text{MSE} = 0.3167 \qquad \text{MAE} = 0.2880$$

2.5 Main Experimental Configuration

Table 1. Main experimental configuration

Setting	Value
Dataset	Weather
Task	Supervised, multivariate-to-multivariate forecasting
Sampling frequency	10 minutes
Variables	21
Input window	336 observations (56 hours)
Main output window / horizon	96 observations (16 hours)
DLinear moving-average kernel	25
Channel configuration	Shared temporal projections across variables (individual=False)
Loss / training objective	MSE
Optimizer	Adam
Initial learning rate	0.0001
Learning-rate schedule	Official DLinear type1 exponential decay (factor 0.5)
Batch size	16
Maximum epochs	10
Early-stopping patience	3 epochs
Checkpoint criterion	Validation MSE
Direct reconstruction seed	2021
Robustness seeds	2021, 2022, 2023

3. Original DLinear Architecture

3.1 Trend-Remainder Decomposition

DLinear first estimates a smooth trend using a centered moving average with kernel size 25:

$$T = \mathrm{MA}_{25}(X), \qquad R = X - T$$

The official implementation repeats the first and last observations before applying the moving average. This preserves the sequence length without adding artificial zero padding. The paper calls (R) the seasonal component. Because it contains all fluctuations left after removing the moving-average trend, we use the more conservative name remainder.

3.2 Direct Linear Forecasting Layers

Two linear layers independently map the complete trend and remainder histories to the complete forecast horizon:

$$\hat{\mathrm{Y}} = W_T T + b_T + W_R R + b_R$$

We reproduce the official shared-projection setting (individual=False). One temporal projection matrix is shared across all variables in each branch. Each variable is nevertheless forecast only from its own decomposed history; there is no direct cross-variable mixing. For input length 336 and horizon 96, the two projections contain 2 x (336 x 96 + 96) = 64,704 trainable parameters. Each projection has a learned 96-dimensional bias vector, so the two biases contribute 192 parameters to the total count.

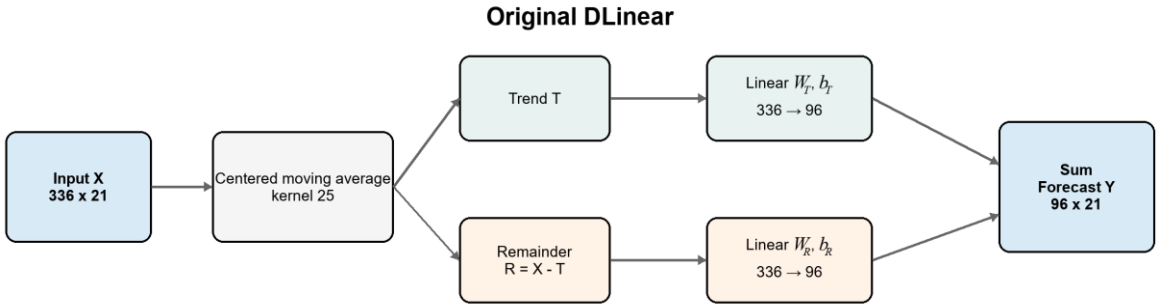


Figure 1. Original DLinear decomposition and direct multi-step forecasting flow.

3.3 Loss Function and Training Objective

The model minimizes elementwise mean squared error:

$$L_{MSE} = \frac{1}{BHC} \sum_{b=1}^{B} \sum_{t=1}^{H} \sum_{c=1}^{C} \left(Y[b,t,c] - \hat{\mathrm{Y}}[b,t,c] \right)^2$$

Here, B is the batch size, H is the forecast horizon, and C is the number of variables. The indices b, t, and c identify a particular input window in the batch, forecast step, and variable, respectively.

Training uses Adam, the paper repository's type1 learning-rate schedule, validation-based early stopping, and the remaining hyperparameters listed in Table 1.

4. DLinear Results Reported in the Paper

Table 2 reproduces the paper's DLinear results on Weather with input length 336. These values are the numerical targets for our reconstruction.

Table 2. DLinear results reported for Weather in the selected paper

Forecast horizon	Real duration	Paper MSE	Paper MAE
96	16 hours	0.176	0.237
192	32 hours	0.220	0.282
336	56 hours	0.265	0.319
720	120 hours	0.323	0.362

Source: Zeng et al. [1], Weather results with look-back length 336.

The paper evaluates forecasting performance using MSE and MAE, where lower values indicate more accurate forecasts. Both errors increase with the forecast horizon: MSE rises from 0.176 at horizon 96 to 0.323 at horizon 720, while MAE rises from 0.237 to 0.362. This indicates that forecasting becomes more difficult over longer output windows. These reported values serve as the reference targets for our reconstruction in Section 5.

5. Reconstruction

5.1 Implementation Fidelity

Our reconstruction follows the official repository in the choices that determine DLinear's behavior: centered moving-average decomposition with repeated-endpoint padding, two direct temporal projections shared across variables (individual=False), MSE training, Adam optimization, the type1 learning-rate schedule, chronological splits, and train-only standardization. The official long-forecasting script sets itr=1, meaning one experiment repetition rather than one training iteration, and the experiment runner fixes seed 2021. We therefore use seed 2021 for the direct paper comparison.

5.2 Evaluation Protocol

After training, the checkpoint with the lowest validation MSE is evaluated once on the test set. The seasonal-naive baseline and DLinear reconstruction use the same split and metrics. Seed 2021 is used for the direct paper comparison; seeds 2022 and 2023 provide additional robustness runs without changing the experimental configuration.

5.3 Comparison with the Paper

Table 3. Paper comparison using reconstruction seed 2021

Horizon	Paper MSE	Reconstruction MSE	Relative Difference	Paper MAE	Reconstruction MAE	Relative Difference
96	0.17600	0.17421	-1.02%	0.23700	0.23333	-1.55%
192	0.22000	0.21598	-1.83%	0.28200	0.27350	-3.01%
336	0.26500	0.26134	-1.38%	0.31900	0.31091	-2.54%
720	0.32300	0.32422	+0.38%	0.36200	0.36249	+0.14%

Negative differences mean our reconstruction obtained lower error. Every value is within 3.01% of the paper, with no systematic degradation. Differences of this size can arise from software versions, random initialization, GPU kernels, and minor evaluation-loader details.

Across seeds 2021, 2022, and 2023, DLinear gives the following mean $\pm$ sample standard deviation results: test MSE 0.17464 ± 0.00052 and MAE 0.23486 ± 0.00174 at horizon 96. This repeated-seed result is our robustness analysis and is not presented as the paper's reported value.

6. Proposed Improvement

6.1 Motivation

Original DLinear applies the same 25-sample moving average to every Weather variable and every input window. This assumes one fixed smoothing scale is appropriate throughout the dataset. Because the 21 variables can exhibit different temporal dynamics, a single scale may smooth some series too little or remove useful short-term structure from others.

We retain DLinear's forecasting layers and modify only its decomposition. Our research question is whether variable and window-specific trend scales can improve forecasting while preserving the simple DLinear backbone.

6.2 Static Per-Variable Multiscale Decomposition

We first test a static variant using moving-average kernels 25, 73, and 145. Kernel 25 retains DLinear's original scale; at ten-minute sampling, the three centered windows span approximately 4, 12, and 24 hours. Each variable learns one input-independent convex mixture:

$$\alpha[c,:] = softmax(a[c,:]), \quad T[b,t,c] = \sum_{m=1}^{3} \alpha[c,m]\mathrm{MA}_{k_m}(X)[b,t,c]$$

Here, $MA_{k_m}(X)$ is the trend obtained using moving-average kernel k_m, and $\alpha[c,m]$ is the learned weight assigned by variable c to that scale. The weights are fixed for each variable and do not change between input windows.

Softmax keeps the weights non-negative and makes them sum to one. This variant tests whether different variables prefer different smoothing scales and adds only 21 × 3 = 63 variable-specific logits.

6.3 Dynamic Per-Variable Multiscale Decomposition

The final variant also adapts the mixture to the current observed window.

First, for each input window b and variable c, it calculates four summary features from observed input values only:

$$f[b,c] = \left[mean_t X[b,t,c], std_t X[b,t,c], X[b,L,c], X[b,L,c] - X[b,1,c]\right]$$

These features are the window mean, standard deviation, last observed value, and the difference between the last and first observed values.

Second, a shared two-layer gate maps these features to a three-element correction vector:

$$d[b,c,:] = W_2 GELU(W_1 f[b,c] + b_1) + b_2$$

The vector $f[b,c]$ has size 4. In the first layer, W_1 has size 8 × 4 and b_1 has size 8, producing a hidden vector of size 8. GELU is applied elementwise, providing a smooth nonlinearity that can model relationships beyond one linear transformation. The second layer uses W_2 of size 3 × 8 and b_2 of size 3 to produce $d[b,c,:]$ of size 3, one correction for each moving-average scale. We chose hidden size 8 as a compact design; the shared gate adds only 67 parameters.

Each variable learns three prior logits $a[c,:]$, which have the same role as the logits in the static variant: they represent its general scale preference. In the dynamic model they are learned jointly with the gate, not copied from the trained static model. Adding the window-dependent correction $d[b,c,:]$ and applying softmax gives three non-negative weights that sum to one:

$$\alpha[b,c,:] = softmax(a[c,:] + d[b,c,:]), \quad T[b,t,c] = \sum_{m=1}^{3} \alpha[b,c,m]\, MA_{k_m}(X)[b,t,c]$$

The second expression forms the trend as a convex combination of the three moving-average outputs, using weights that can differ between input windows and variables.

The remainder is $R = X - T$, and the original DLinear projections forecast both components.

When $d[b, c, :]$ is close to zero, the method behaves like a static per-variable mixture; otherwise, its weights adapt to the current window. The gate uses neither target nor future values. The prior logits and gate output layer are initialized to zero, so all scales start equally weighted and the dynamic correction initially vanishes. During training, the prior learns general variable-specific preferences while the gate learns window-specific adjustments.

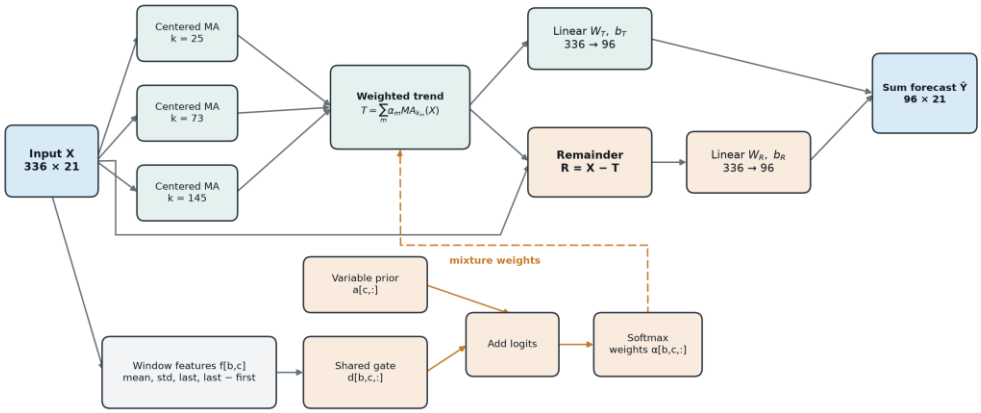


Figure 2. Proposed dynamic per-variable multiscale decomposition while retaining DLinear's forecasting layers.

6.4 Parameter Count and Training Strategy

Compared with original DLinear, the static model adds 63 variable-specific logits. The dynamic model retains these 63 prior logits and adds 67 parameters in the shared gate, giving a total increase of 130 parameters. The architecture of the original trend and remainder forecasting projections remains unchanged.

Table 4. Model parameter counts for input length 336 and forecast horizon 96

Model	Parameters	Increase from DLinear
Original DLinear	64,704	-
Static per-variable multiscale DLinear	64,767	+63 (+0.097%)
Dynamic per-variable multiscale DLinear	64,834	+130 (+0.201%)

The improved models use the same data split, scaling, optimizer, loss, learning-rate schedule, batch size, early-stopping rule, and paired seeds as the reconstruction. For each seed, DLinear and the improved model begin with identical randomly initialized trend and remainder projection weights; no trained DLinear checkpoint is transferred. The new multiscale parameters are initialized separately. Each model is then trained independently for up to ten epochs, with all parameters optimized jointly. Architecture selection uses validation MSE before the final test evaluation.

7. Results

7.1 Main Paper-Reconstruction-Improvement Comparison

Table 5 gives the assignment's required comparison at the main 96-step horizon. The paper row is the published DLinear value, while our reconstruction and improved model report mean and sample standard deviation across three paired seeds.

Table 5. Main comparison at forecast horizon 96

Model	Test MSE	Test MAE
Paper DLinear	0.17600	0.23700
Our DLinear reconstruction	0.17464 ± 0.00052	0.23486 ± 0.00174
Dynamic per-variable multiscale DLinear	0.16540 ± 0.00045	0.22492 ± 0.00137

Relative to its paired DLinear runs, the dynamic model reduces mean MSE by 5.29% and MAE by 4.23%. It obtains lower error for every paired seed on both metrics.

7.2 Ablation Study

Table 6. Ablation results at forecast horizon 96

Model	Test MSE (mean ± SD)	Test MAE (mean ± SD)	MSE reduction vs DLinear	MAE reduction vs DLinear
DLinear	0.17464 ± 0.00052	0.23486 ± 0.00174	-	-
Static per-variable multiscale	0.16739 ± 0.00044	0.22910 ± 0.00114	4.15%	2.45%
Dynamic per-variable multiscale	0.16540 ± 0.00045	0.22492 ± 0.00137	5.29%	4.23%

The static model's improvement supports the hypothesis that different variables benefit from different decomposition scales. The additional gain from the dynamic model suggests that the useful scale also depends on the current input window. Dynamic adaptation reduces MSE by approximately 1.19% and MAE by 1.82% relative to the static model.

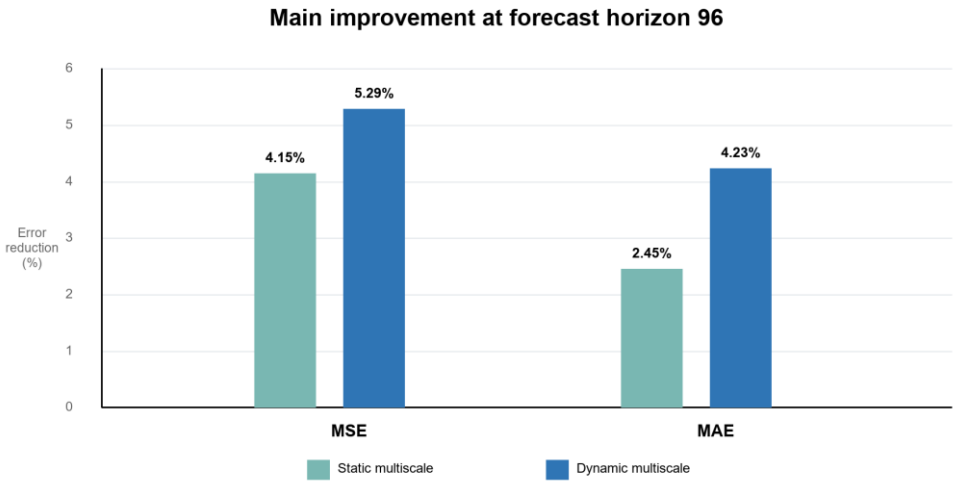


Figure 3. Mean paired error reduction at the main 96-step forecast horizon.

7.3 Multi-Horizon Robustness

We repeat the paired three-seed comparison at all four horizons used by the paper while keeping input length 336. The same dynamic architecture is used without horizon-specific redesign. Values are reported as mean ± sample standard deviation across seeds 2021, 2022, and 2023, and relative reductions are calculated from the corresponding mean errors.

Table 7. Three-seed multi-horizon comparison

Horizon	DLinear MSE	Dynamic MSE	MSE reduction	DLinear MAE	Dynamic MAE	MAE reduction
96	0.17464 ± 0.00052	0.16540 ± 0.00045	5.29%	0.23486 ± 0.00174	0.22492 ± 0.00137	4.23%
192	0.21570 ± 0.00029	0.20875 ± 0.00072	3.22%	0.27270 ± 0.00082	0.26537 ± 0.00175	2.69%
336	0.26297 ± 0.00142	0.25730 ± 0.00214	2.16%	0.31445 ± 0.00313	0.30870 ± 0.00422	1.83%
720	0.32444 ± 0.00069	0.31983 ± 0.00109	1.42%	0.36299 ± 0.00125	0.35737 ± 0.00151	1.55%

The dynamic model improves both metrics for every seed at every evaluated horizon. The relative benefit remains positive as forecasting becomes more difficult, although it generally decreases with horizon length.

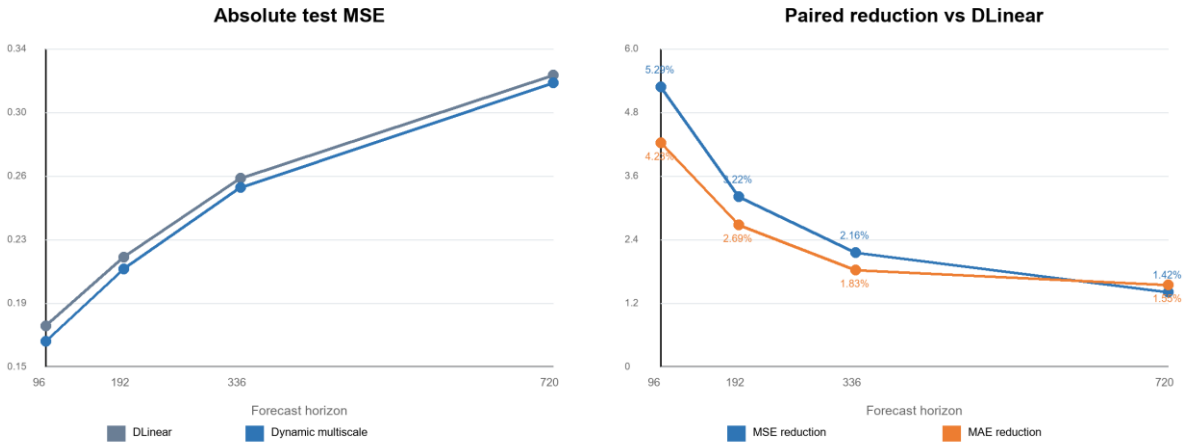


Figure 4. Absolute MSE and paired relative reductions across all evaluated horizons.

8. Discussion, Robustness and Limitations

8.1 What Worked

The reconstruction is close enough to the paper to provide a credible experimental baseline. The ablation also gives a clear progression: variable-specific scales improve DLinear, and input-window adaptation improves further. The result is not caused by one favorable seed because the dynamic model wins every paired comparison. It is also parameter-efficient, adding only 130 parameters to the 64,704 parameter baseline.

8.2 Limitations

The relative improvement decreases with forecast length: MSE reduction falls from 5.29% at horizon 96 to 1.42% at horizon 720. This suggests that adaptive decomposition mainly improves the representation of the observed input, while uncertainty and other limitations dominate at longer horizons.

The small parameter increase does not imply negligible computational cost. Compared with DLinear, the dynamic model computes three moving averages instead of one, four summary features for every window-variable pair, and a small $4 \rightarrow 8 \rightarrow 3$ feedforward gate followed by softmax. Although the gate is lightweight, these additional operations increase runtime. We do not report runtime as a formal efficiency result because early stopping selected different epoch counts, so the runs do not provide a controlled speed comparison.

Finally, the experiments cover one dataset and three seeds. Multi-horizon consistency strengthens the conclusion on Weather, but it does not prove that the same kernels and gate will improve every time series.

8.3 What We Learned

We learned that decomposition is not merely a preprocessing step: its scale can become a small learnable part of the forecasting model. Testing the static version before adding a dynamic gate was useful because it separates two effects: different variables prefer different scales, and those preferences also change with the observed window.

Additionally, we also learned that careful temporal evaluation is essential for determining whether an apparent improvement is genuine. Using train-only scaling, keeping forecast targets within their assigned partitions, selecting models only on validation data, comparing paired seeds, and evaluating the test set only after model selection helped us avoid temporal leakage and test-driven tuning.

9. Conclusions

In this project, we faithfully reconstructed DLinear on the Weather benchmark and proposed dynamic per-variable multiscale DLinear. The method replaces one fixed moving-average trend with an input-dependent convex mixture of short, medium, and daily scale trends while preserving DLinear's direct linear forecasting layers.

At horizon 96, the proposed model reduces mean test MSE by 5.29% and mean test MAE by 4.23% across three paired seeds while adding only 0.201% more parameters. It also improves both metrics for every paired seed at horizons 192, 336, and 720. Within the evaluated task, adaptive multiscale decomposition is therefore a meaningful, robust, and interpretable improvement to DLinear.

References

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